KUDZAI P. MATIZIROFA

AI Engineer · LLMs & Voice AI · Machine Learning · Cloud AI Deployment

Harare, Zimbabwe | +263 78 197 4149 | kudzaiprichard@gmail.com

linkedin.com/in/kudzai-prichard | github.com/kudzaiprichard | huggingface.co/Teleagents | prichard.co.zw

SUMMARY

AI Engineer with 2+ years shipping production ML models, LLM-powered agents, OCR and speech systems through Python APIs on AWS and Azure. Co-founder of **Teleagents**, named **Champion at the POTRAZ Innovation Expo 2026**. Built document-AI and banking systems for EcoCash, CABS, Stanbic, BancABC and Old Mutual.

EXPERIENCE

Teleagents | **Co-Founder & AI Engineer** · *voice AI agents for African languages* — Harare | *2025 – Present*

- **Built and open-sourced Zimbabwe All Speech**: 1,678+ hours and 358K clips across Shona, Ndebele and Ndau. It merges 7 sources into one schema with per-clip quality scores, and uses pseudo-labelling to grow 15K human-labelled clips 6x. Dataset
- Trained **in-house ASR and TTS models** for Zimbabwean languages on SageMaker and EC2 GPUs, fed by an S3 + Spark (EMR) pipeline for segmentation and quality tiering, giving native Shona voice that commercial providers lack.
- Architected a **self-hosted real-time voice agent platform** (Python, FastAPI, Next.js, telephony) with streaming STT → LLM → TTS and **RAG over a Redis vector database**, so agents answer from each client's own knowledge. Auto-scaled on AWS ECS.

Sybrin Imaging Solutions | **AI Systems Engineer** · *enterprise fintech* — Harare | *Aug 2024 – Jul 2026*

- **Shipped a production document-AI (OCR) platform for EcoCash**, CABS and Stanbic (C#, .NET, Azure, OpenAI, Redis), processing tens of thousands of documents a month. Migrated it from Azure Document Intelligence to a more accurate AI engine.
- **Built 2 production AI agents for BancABC**: an RTGS payment-instruction processor that ended manual transaction keying, and an HR pipeline for CV intake, NLP resume parsing and candidate scoring. Trained custom computer-vision models (TensorFlow, OpenCV) that extract faces and signatures from scanned forms.
- **Designed the CABS Internet Banking API from scratch** (C#, .NET, MySQL), with transaction endpoints, a background indexer, audit logs and OAuth/JWT + MFA for regulatory compliance. Also built a Telegraphic Transfer service with mobile APIs.
- Closed 100+ tickets on Old Mutual's microservices platform (6 countries) and ran client demos; built a low-code UI layer adopted by 4 tier-1 clients.

OurAfrica | **Co-Founder & Engineer** · *offline-first e-learning* — Harare | *2025 – Present*

- Built an offline-first learning platform (web, desktop, Android) on a Spring Boot API hosted on AWS.

Replica Systems | **Software Engineer Intern** — Harare | *Jan 2022 – Jun 2022*

- Took a construction site-management platform from an empty repo to production (Spring Boot, PostgreSQL, Angular), with JWT auth, RBAC, rate limiting and CI/CD to AWS and Vercel.

HONORS & AWARDS

- **Champion, POTRAZ Innovation Expo & Conference 2026** (Bulawayo, Oct 2026) for "TeleAgents: AI Voice Agents for Zimbabwe". Also recognised as Innovator by the POTRAZ Director-General (an ITU Acceleration Centres Network member).

SELECTED PROJECTS

- **Metis** (PyTorch, FastAPI, Neo4j) – Thesis. Type-2 diabetes treatment recommender that learns from outcomes, built on a Neural Thompson Sampling contextual bandit with hard safety-gate overrides and Neo4j similar-patient retrieval.
- **AURA** (FastAPI, Scikit-Learn, Next.js) – Phishing and fraud detector that keeps learning from analyst feedback, with drift monitoring, a hot-swap model registry and a Gmail extension. GitHub
- **Loan Default Prediction** (LightGBM, CatBoost, Optuna) – Deep Learning IndabaX Zimbabwe 2026. Adversarial validation found 91.5% train/test separability. Built a shift-robust ensemble tuned over 50+ experiments (0.6840 public LB). GitHub
- **Passport Verification API** (Python, Scikit-Learn, Flask) – ML API that detects forged passports for identity (KYC) checks.

SKILLS

Languages: Python, C#, Java, TypeScript, SQL

AI / ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, LLMs, RAG / vector search, AI agents, ASR / TTS, NLP, computer vision, OCR

Cloud AI: AWS SageMaker, Azure OpenAI, Azure Document Intelligence, OpenAI API, Hugging Face

APIs & Data: FastAPI, Flask, .NET, Spring Boot, REST microservices, OAuth / JWT / MFA, PostgreSQL, MySQL, MongoDB, Redis, Spark

Cloud & DevOps: AWS (EC2, ECS, ECR, S3, EMR, Glue, Lambda, RDS, CloudWatch), Azure, Docker, CI/CD, Linux | React, Next.js

EDUCATION

B.Sc. Honours, Artificial Intelligence & Machine Learning – University of Zimbabwe · *Upper Second Class (2:1)* — *2022 – 2026*

Diploma & Certificate in Software Engineering – University of Zimbabwe — *2019 – 2022*

Level 4 & 5 Diplomas in Computer Engineering – ABMA Education, UK — *2018 – 2019*